

Confirmation (Progress Review 1) Report

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1 Administrative Details

- **Project title:** Modelling spin-crossover materials with machine-learned force fields
- **Advisors:** Dr Carla Verdi (Principal Advisor); Professor Ben Powell (Associate Advisor)
- **Starting date:** 1 January 2025
- Funded by Australian Government Research Training Program Scholarship.

2 Introduction and Aims

A transition metal ion surrounded by non-metallic ligands can form a coordination complex with the capacity to exist in two different electronic ground-states - one low spin (LS) and one high spin (HS) [1]. Given external perturbations in temperature [2, 3], pressure [4], applied magnetic fields [5], or light irradiation [6, 7], the system can interconvert between the two states, in a process known as spin-crossover (SCO).

Other physicochemical changes accompany the crossover, inviting a range of exciting potential applications. For example, changes in magnetism [8] and colour [9, 10] make SCO compounds promising molecular switches [11], with proposed integration into sensors [12, 13] and high-density memory devices [14, 15]. However, advancing the viability of any of these applications requires that SCO can be triggered reversibly under ambient conditions [15]. It is imperative that a deep understanding of collective phenomena influencing SCO energetics and dynamics in the solid state is developed accordingly.

To accelerate the design and discovery of candidate SCO materials, it would be desirable to calculate their electronic and structural properties purely from first-principles (*ab initio*). The most popular way to approach this for crystal structures is via density functional theory (DFT) [16], as it is much cheaper computationally than other *ab initio* electronic structure methods (especially when applied to large unit cells with hundreds of atoms) [17, 18]. Despite this, choosing an optimal DFT method to model SCO systems is no trivial task. Even after extensive benchmarking studies, there is no widespread agreement on best practice for these materials, largely due to the challenge of accurately treating the strongly correlated valence electrons of the transition metal [18, 19, 20, 21].

While a properly-tuned DFT method can give accurate optimised geometries and static energy calculations in both spin states, the approach does not directly extend as well to studies of SCO *dynamics*. Molecular dynamics (MD) simulations take interatomic forces as input to progressively update the atomic positions of a system. One can then track its evolution as a function of temperature. Such forces may indeed be calculated via first-principles - but, with timesteps typically on the order of femtoseconds, this quickly becomes prohibitively expensive to iteratively calculate for large systems at longer timescales [22, 23, 24]. It is more common to instead run MD simulations with *force fields* - that is, parametrisations of the potential energy [25].

Developing accurate force fields manually is time-consuming and challenging - often, some properties are accurate, while others fail - but they have been developed by other groups to model qualitative SCO dynamics [23]. Excitingly, constructing new and highly accurate force fields has become much easier with recent advances in machine learning [26]. Without any knowledge of chemistry, machine-learned force fields (MLFFs) learn the relationship between atomic structures and DFT energies. Once trained, the MLFF can be applied to MD simulations, achieving DFT-level accuracy at a fraction of the computational cost [27, 28, 29]. This forms the central goal of my project - to model the dynamics of thermal SCO with *bespoke MLFFs*. We first benchmark the methodology with a prototypical SCO material, $[\text{Fe}(\text{ptz})_6](\text{BF}_4)_2$ (ptz = 1-propyltetrazole), chosen for its well-characterised thermal transition [9]. In the future, we hope to construct MLFFs for newly synthesised SCO materials, aiding experimentalists in the screening and design of multifunctional materials at targeted transition temperatures.

3 Background

3.1 Spin-crossover systems

SCO was first discovered almost a century ago in coordination complexes with Fe(III) metal centres [2]. The phenomenon may generally be observed for transition metal ions with electronic configurations d^4 to d^7 , corresponding to the number of outer electrons in the partially-filled d shell. To explain how crossover occurs, we refer to Fe(II) octahedral complexes (with a d^6 electronic configuration), as these have been studied most extensively [19, 30].

In free space, the six d electrons of an Fe(II) ion can occupy any of the five possible d orbitals. This degeneracy is broken when the ion bonds to six ligands and forms an octahedral complex (as in Figure 1). Defining the Cartesian x, y and z axes to lie along the metal-ligand bond directions, the two d_{z^2} and $d_{x^2-y^2}$ orbitals point directly along these axes and therefore experience greater electrostatic repulsion than the three non-axial orbitals. The five d orbitals are then split by the ligand field into the higher energy, two-fold degenerate e_g set, and the lower energy, three-fold degenerate t_{2g} set, following standard notation from group theory.

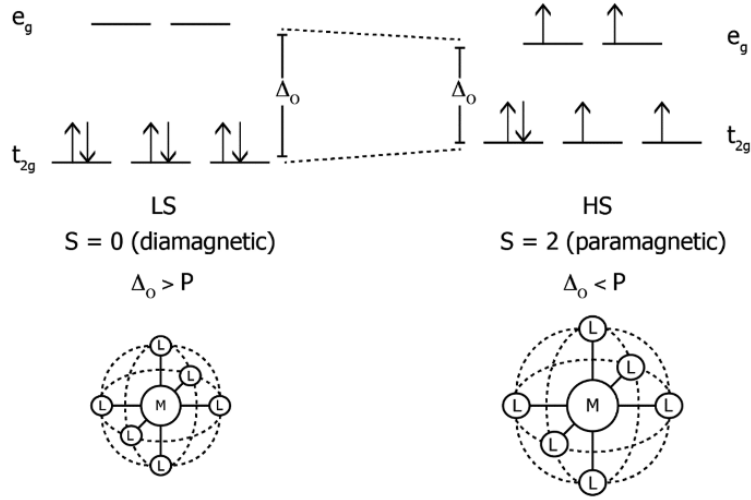


Figure 1: Schematic energy level diagram showing the two states of an octahedral d^6 SCO material, depending on the magnitude of the ligand field splitting energy, Δ_0 . P is the spin pairing energy. This figure is reproduced from Vallone *et al.* [31].

Figure 1 shows the conditions which make either spin state energetically favourable, in terms of the ligand field splitting energy, Δ_0 , and the spin pairing energy, P . Under a strong ligand field, it is more costly to promote an electron to the higher energy e_g orbitals than to pair it up in the lower energy orbitals. This keeps all six electrons in the t_{2g} orbitals, and the Fe(II) ion in the diamagnetic LS state, with total spin angular momentum of 0. In a weaker ligand field (relative to P), it may be more favourable to leave electrons unpaired, placing them across high and low energy orbitals instead via Hund's rules [32]. The four unpaired electrons in the resulting HS state make it paramagnetic, with a total spin angular momentum of 2. If Δ_0 is comparable to P , then both spin states become thermally accessible - the material can undergo SCO.

A key parameter for characterising and designing SCO systems is the transition temperature, $T_{1/2}$ [23, 33, 34]. The transition from low to high spin occurs with increasing temperature, and, at $T_{1/2}$, there are an equal number of LS and HS molecules in the system. The transition can be monitored with magnetic susceptibility measurements (due to the magnetic changes between the LS and HS states explained above), and an experimental value for $T_{1/2}$ extracted accordingly [1].

It is also possible to derive a neat expression of $T_{1/2}$ to make some *predictions*. This is clearly critical for any functional applications that require spin state switching at room temperature [15]. We start by writing the Gibbs free energy difference (ΔG) between the two states as

$$\Delta G = G_{\text{HS}} - G_{\text{LS}} = \Delta H - T\Delta S, \quad (1)$$

where ΔH is the enthalpy change $H_{\text{HS}} - H_{\text{LS}}$, T is the temperature, and ΔS is the entropy change $S_{\text{HS}} - S_{\text{LS}}$. At thermodynamic equilibrium, the free energies of the LS and HS states are equal, and $\Delta G = 0$ [34]. It then follows

from Equation (1) that

$$0 = \Delta H - T\Delta S \implies T_{1/2} = \frac{\Delta H}{\Delta S}. \quad (2)$$

So, the transition temperature of SCO can be calculated from just the enthalpy difference ΔH and the entropy difference ΔS between the two states. To compute these quantities, theoretical calculations should include both electronic and vibrational contributions to the enthalpy, $\Delta H = \Delta H^{\text{elec}} + \Delta H^{\text{vib}}$, and the entropy, $\Delta S = \Delta S^{\text{elec}} + \Delta S^{\text{vib}}$ [18, 33].

The electronic terms H^{elec} and S^{elec} are temperature-independent, to good approximation [35]. In practice, H^{elec} for a given state is directly obtained from electronic structure methods [23], while ΔS^{elec} is a constant value only depending on the change in spin multiplicity between HS and LS. To calculate this, we compare the change in total orbital and spin momenta $L_{\text{LS/HS}}$ and $S_{\text{LS/HS}}$ by

$$\Delta S^{\text{elec}} = \Delta S^{\text{orb}} + \Delta S^{\text{spin}} = k_B \ln \left(\frac{2L_{\text{HS}} + 1}{2L_{\text{LS}} + 1} \right) + k_B \ln \left(\frac{2S_{\text{HS}} + 1}{2S_{\text{LS}} + 1} \right). \quad (3)$$

For the d^6 octahedral configuration examined above, this evaluates to $k_B \ln(3) + k_B \ln(5) = k_B \ln(15) = 0.23 \text{ meV K}^{-1}$ per metal centre [36].

Even at 0 K, though, atoms undergo small oscillations about their equilibrium positions. These can be decomposed into a set of normal modes, which, under the quantum harmonic oscillator assumption, vibrate independently of each other. At finite temperature, these modes (with frequencies ω) can be thermally populated, giving rise to the temperature-dependent H^{vib} [18]. For a single SCO molecule in a given spin state,

$$H^{\text{vib}}(T) = \sum_{i=1}^{3N-6} \left(\frac{\hbar\omega_i}{2} + \frac{\hbar\omega_i \exp(-\frac{\hbar\omega_i}{k_B T})}{1 - \exp(-\frac{\hbar\omega_i}{k_B T})} \right), \quad (4)$$

where N is the number of atoms in the molecule, $\hbar$ is the reduced Planck constant, and k_B is Boltzmann's constant. The first term in Equation (4) corresponds to the zero-point energy (ZPE) arising from molecular vibrations at 0 K. Finally, the vibrational entropy for a single SCO molecule is [37]

$$S^{\text{vib}}(T) = \sum_{i=1}^{3N-6} \left(\frac{\hbar\omega_i}{2T \tanh(\frac{\hbar\omega_i}{2k_B T})} - k_B \ln \left(2 \sinh \left(\frac{\hbar\omega_i}{2k_B T} \right) \right) \right). \quad (5)$$

Typically, $\Delta S^{\text{vib}} > \Delta S^{\text{elec}}$ - in Fe(II) compounds, it is between $2\text{-}4\times$ bigger [38].

Key structure-property relationships are also coupled to the vibrational modes of the SCO material. To account for occupation of the antibonding e_g orbitals in the HS state, the average bond length between the metal and ligands increases by $\sim 10\%$ in Fe(II) complexes [36, 38]. The bond weakening has flow-on effects on the vibrational spectra of the material, with reduced low-frequency metal-ligand modes in the HS state [37].

3.2 *Ab initio* methodologies: energetics and dynamics

From our discussion above, it is clear that, within the harmonic approximation, predictions of the transition temperature, $T_{1/2}$, are possible without necessarily simulating any SCO dynamics. All that is required are reliable calculations of the energy differences between the two spin states and their vibrational spectra. Simulations of crossover itself, though, are also desirable - temperature effects can be directly treated [22], and the sharpness of the transition can be studied to infer viability for switching applications. An introduction to the methods of choice for modelling energetics (DFT) and dynamics (MD) from first principles is given below, with a brief discussion of how they can be used to treat SCO.

3.2.1 Density functional theory

To completely describe the ground-state energy of a collection of atoms, one would need to solve the time-independent Schrödinger equation,

$$\hat{H}\Psi = E\Psi, \quad (6)$$

where $\hat{H}$ is the Hamiltonian operator and Ψ comprises the set of solutions for the Hamiltonian. Each solution Ψ_n has an associated eigenvalue, E_n , a real number that satisfies Equation (6). But, beyond trivial systems and a

few special cases, it is not possible to exactly determine Ψ_0 corresponding to the ground-state energy E_0 . This is because $\hat{H}$ has an electron-electron repulsion term, which makes this a many-body problem.

The crux of DFT is that it approximates a solution to the many-body Schrödinger equation via the electron density $n(\mathbf{r})$ - a physical observable that depends only on 3 spatial coordinates, massively simplifying the problem [17]. Hohenberg and Kohn proved that the energy of the system, from the Schrödinger equation, is a unique *functional* of $n(\mathbf{r})$, $E[n(\mathbf{r})]$ [39]. A functional takes a function as its input and maps it to a scalar. Hohenberg and Kohn also proved that the electron density $n(\mathbf{r})$ which minimises E , $n_0(\mathbf{r})$, is the *true ground-state electron density* corresponding to a full solution of Equation (6) [39]. One could imagine varying $n(\mathbf{r})$ until the energy from the functional is minimised. Unfortunately, the exact form of the functional $E[n(\mathbf{r})]$ is unknown, but it can be split into the analytic and leftover parts,

$$E[n(\mathbf{r})] = E_{\text{known}}[n(\mathbf{r})] + E_{\text{XC}}[n(\mathbf{r})] \quad (7)$$

where E_{known} treats kinetic energies and direct Coulomb interactions, and E_{XC} is the unknown exchange-correlation functional. Many approximations of E_{XC} have been formulated, with varying levels of complexity and parametrisation [40].

Ultimately, the choice of functional for a given material should be guided by benchmarking studies. Even when we limit our search to a fixed metal centre, there is still no agreement on a single optimal functional for SCO systems. A selection of hybrid functionals (parametrised from experiment) seem to perform best for ΔH predictions in Fe(II) complexes [19, 20], but they are computationally expensive to use and can no longer be considered completely *ab initio* [41].

Strong correlations

Strongly-correlated systems are characterised by highly localised electrons. Transition metals with open d orbitals are quintessential examples of such a system [42] - of course, they also generate the key physics of SCO. It is troublesome, then, that DFT has well-known problems computing accurate energies of strongly-correlated systems [43]. How can this be, and what is the workaround?

The source of the problem is the systematic self-interaction error. In the exact functional posited by Hohenberg and Kohn, the Coulomb term in E_{known} includes an unphysical interaction of $n(\mathbf{r})$ with itself, which would be exactly cancelled by E_{XC} if its form was known. Of course, since E_{XC} is only approximate, there is an erroneous self-repulsion which artificially favours electron delocalisation, and misrepresents the strong correlations that underpin transition metals [19, 44].

An approximate correction to this over-delocalisation of d electrons involves adding an on-site interaction term to a pure DFT functional, in a popular method called DFT+U [45, 46]. The additive U term is treated within a mean-field approximation, acting on the d orbitals to penalise partial occupation and effectively increase electron localisation [47]. Many groups have used DFT+U methods to model SCO energetics, as they reduce the self-interaction error while being much cheaper than hybrid functionals [18, 22, 21, 47]. The choice of U is material-specific and requires benchmarking [33, 47].

Phonons

Finally, a note on vibrational contributions to H and S in the solid-state. The fundamental input for a DFT calculation is a periodic unit cell, with atomic positions repeated throughout space to represent an extended crystal lattice. This means the vibrational modes which contribute thermal energy to SCO *crystals* are no longer localised (as in Equations (4) and (5)), but are phonons - collective modes which oscillate over the whole lattice [48].

Conceptually, obtaining the *molecular* vibrational normal modes is analogous to calculating the *phonon* modes, with thermal occupation following Bose-Einstein statistics [48]. To calculate the phonon modes, *ab initio* forces are calculated for versions of the unit cell in which every atom is slightly displaced from its equilibrium position. From this, a Hessian matrix for the crystal can be written and mass-weighted to form the *dynamical* matrix, whose diagonalisation yields the phonon frequencies [49].

From the phonons, one can calculate thermodynamic parameters like $\Delta G(T)$, which gives a way to evaluate $T_{1/2}$ as the crossing point where $\Delta G = 0$. The formulas are the same as for a single molecule, except the sums iterate over all phonon modes. For a more accurate treatment of free energies, one can further apply a quasi-harmonic approximation, which incorporates anharmonicity by calculating the phonons' volume-dependence. This requires multiple phonon calculations at different volumes [50]. For SCO systems, computing phonon frequencies in the crystalline phase is rarely done [33] - the method above is computationally expensive, with DFT forces

from finite displacements in $\pm x$, y , and z computed for every atom in a large unit cell, over a range of different volumes. In solid-state modelling, the vibrational modes from single molecules are often taken as an approximation instead [23], or tuned to match experimental ΔS [18]. The benefits of using a force field trained on DFT data for (otherwise, prohibitively expensive) quasi-harmonic phonon calculations then become apparent.

3.2.2 Molecular dynamics simulations

The motivation for simulating SCO dynamics is apparent by comparing the indicative thermal transition curves in Figure 2. All three processes share the same $T_{1/2}$, but there are differences in *how* the systems switch with changing temperature. Figure 2(a) and 2(b) show gradual and sharper transitions, with the first being more typical for SCO in solution and the second more so for solids (due to strong elastic interactions between molecules in the lattice) [6]. Figure 2(c) shows a hysteresis loop, which means the transition occurs at slightly different temperatures depending on whether the system is being heated or cooled. One aim for designing multifunctional SCO materials is to engineer a wide hysteresis loop, so that a particular spin state is stable for small fluctuations in temperature [51].

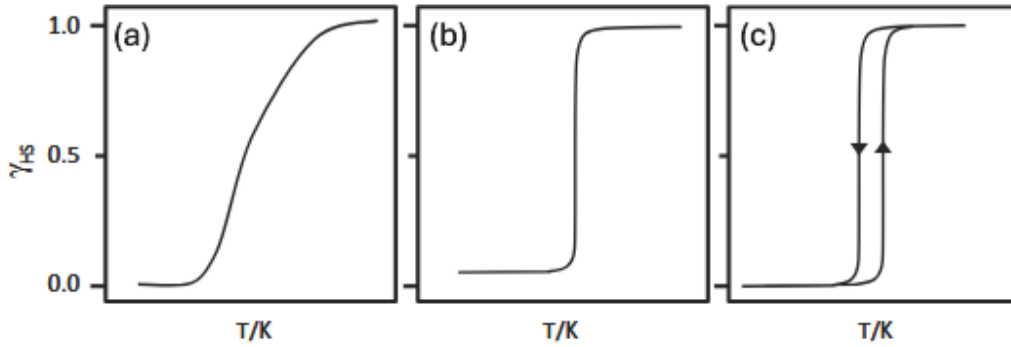


Figure 2: Schematic representation of different SCO curves, each sharing the same $T_{1/2}$, where the proportion of HS to LS states (γ_{HS}) is 50/50. Transitions are (a) gradual, (b) abrupt, and (c) with hysteresis. This figure is reproduced from Cirera and Ruiz [33].

Let us explore the fundamentals of classical MD simulations. MD requires the forces $\mathbf{F}_i$ acting on each atom i to be calculated, so that the atomic positions $\mathbf{r}_i$ can be updated accordingly for the next time step. The workhorse determining these updates is the Verlet algorithm [52],

$$\mathbf{r}_i(t + \Delta t) \approx 2\mathbf{r}_i(t) - \mathbf{r}_i(t - \Delta t) + \frac{\mathbf{F}_i(t)}{m_i} \Delta t^2, \quad (8)$$

which is derived from the central differences of a Taylor expansion about t , from integrating Newton’s equations of motion. With a sufficiently small timestep Δt , this simple expression gives an accurate approximation of the true trajectory [24]. Since we are interested in the trajectories at particular temperatures, we choose a thermodynamic ensemble to couple our system to. Practically, this means we apply an additional temperature control algorithm while iteratively updating the atomic positions. This temperature control is achieved by randomly adjusting some of the atomic velocities. The resulting kinetic energy distribution follows Maxwell-Boltzmann statistics at the target temperature, so T fluctuates around an equilibrium value [53].

Unsurprisingly, the main limitation of *ab initio* MD (AIMD) is its computational cost, stemming from long simulation times and large system sizes. To illustrate, many crystals have phonon modes with frequencies in the THz regime, including SCO materials [54]. One vibration in this regime has a period of around 100 fs, so multiple iterations at timesteps of ~ 10 fs or less are required to describe these trajectories accurately [24]. Furthermore, the spin state switching speed in solids is reported on the nano- to microsecond scale [55], demanding $10^5 - 10^8$ MD steps at least to track the switch (even without temperature equilibration)! This is a considerable investment of resources. The work of Tarafder *et al.* is one of few studies that have used AIMD to simulate SCO, doing so with a (relatively) small unit cell of 38 atoms [22]. Cirera *et al.* combined MD with a Monte Carlo scheme, allowing instantaneous spin state switches for a given configuration of atoms if deemed energetically favourable. They developed a custom force field parametrised on hybrid DFT calculations to circumvent the resource demands of AIMD. While the sharp experimental crossover was recovered, $T_{1/2}$ was overestimated by $\sim 30\%$, and the

discontinuous Monte Carlo dynamics prevent the study of time-dependent switching dynamics [23]. I also reiterate that developing these potentials is time-consuming and requires considerable expertise in itself [56].

3.3 Machine-learned force fields

Machine learning tools have immense potential to revolutionise computational chemistry [26]. In this project, we utilise it with the aim of efficiently producing system-specific MLFFs that can accurately model bulk properties of SCO. This efficiency comes from reducing the need for human intervention, both in the process of fitting to DFT data and sampling said training data.

The process of creating an MLFF is depicted in Figure 3, which approximately structures the rest of this Subsection. First, using an appropriately-chosen DFT scheme, the energies, forces, and stresses are calculated for a set of training structures. Next, the atomic structures are ‘translated’ to machine-readable form. Finally, the statistical relationship between structure and potential energy is learnt, producing a force field that can be used out-of-the-box to perform tasks previously too expensive to do *ab initio*. Some examples already discussed in Section 3.2 are calculating the phonon density of states in complex SCO crystals, and running MD for large systems.

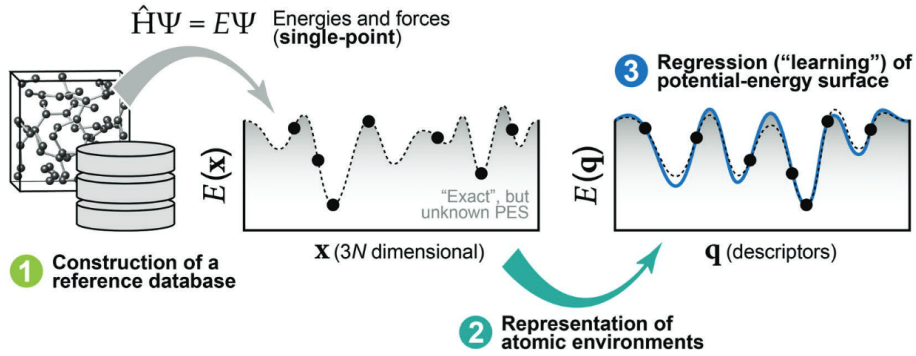


Figure 3: Pictorial overview of the MLFF construction process. This figure is reproduced from Deringer *et al.* [57].

The MLFF can progressively learn as the training dataset is being built up from an AIMD run. At each MD step, either the current MLFF is used to calculate energies and forces, or, if the predicted uncertainties from the force field exceed a certain threshold, a DFT calculation is done instead, and the structure is potentially added to the training set. In this way, up to 98% of the expensive AIMD steps eventually get skipped while assembling the dataset [27]. This ‘on-the-fly’ learning scheme has been integrated into the popular periodic DFT code, Vienna *Ab initio* Simulation Package (VASP) [58], which has been used to perform all of the DFT calculations and MLFF training for this project.

As in step 2 of Figure 3, the local environment around each atom i , in every reference structure N , is encoded in machine-readable *descriptors*. These are rotationally-invariant two- and three-body atomic distribution functions that are expressed as a combination of radial coefficients $\mathbf{x}_i^{(2)}$ and angular coefficients $\mathbf{x}_i^{(3)}$. So, for each atom i , $\mathbf{x}_i = (\mathbf{x}_i^{(2)}, \mathbf{x}_i^{(3)})$ - the 3N-dimensional structural data is mapped to a lower-dimensional descriptor space [59].

For the final step, the descriptors for each pair of atoms $\langle i, j \rangle$ are input into kernel functions $K(\mathbf{x}_i, \mathbf{x}_j)$. Kernel functions quantify the similarity between local environments, with an output of 0 meaning no similarity and an output of 1 meaning they are identical. The DFT energies are then interpolated as linear combinations of these kernel evaluations, with weights w . The same is done for DFT forces and stresses, but with linear combinations of *derivatives* of K with respect to atomic coordinates and lattice vectors, respectively. The coefficients $\mathbf{w}$ are obtained by solving the regression problem

$$\mathbf{Y} = \Phi \mathbf{w}, \quad (9)$$

where $\mathbf{Y}$ is a supervector of all DFT data and Φ is the design matrix of kernels and kernel derivatives. The sum of squared differences between DFT and MLFF predictions is minimised, and the trained MLFF is ready for use [27, 56, 57].

Another way to train MLFFs, as in step 3, is using neural networks (NNs) [57]. To reach the same accuracy as kernel regression methods, NNs generally require more training data, which can be an issue if the source of said data is AIMD [26]. Even so, the approach has been used successfully for small transition metal systems - just last

year, Romano *et al.* successfully modelled the dissociation of water across magnetic surfaces via MLFF-assisted MD [29].

4 Approach, Methods and Timeline

The main aim of this project is to model SCO dynamics in the solid-state with bespoke MLFFs. To do this, I must select a reliable *ab initio* method to provide reference data in both spin states. Next, an MLFF should be trained on-the-fly, before being used to unlock thermal SCO simulations at realistic timescales. The combination of Dr Verdi’s expertise in DFT/MLFF methodologies (particularly as implemented in VASP) with Professor Powell’s experience modelling SCO means that my advisory team has the key skills required for this project.

I foresee the project being split into three steps. The first step is a proof of concept - I must verify that it is possible to create an MLFF which can model properties of solid-state SCO. The second step involves testing the methodology against materials with more exotic transitions. The third step will involve applying the methodology to unpublished materials. I aim for each step to provide a paper and chapter of my thesis.

Step 1: Benchmarking SCO MLFFs on a well-characterised material

The benchmarking material was selected as $[\text{Fe}(\text{ptz})_6](\text{BF}_4)_2$ (ptz = 1-propyltetrazole), a Fe(II) complex with six octahedral Fe-N bonds generating SCO. It has been studied since the 1980s, and its crystallographic, vibrational, and energetic properties are well-known [9, 60]. It also has a sharp thermal crossover at $T_{1/2} = 135$ K.

The DFT+U method was selected to treat the self-interaction error from correlated d electrons, with the (relatively cheap) PBE functional [61]. Separate LS and HS MLFFs were trained on-the-fly, but an erroneous choice of $U = 1.9$ eV, tuned to ΔH^{elec} *without* any vibrational contributions, was made. I redid the benchmarking for U against the direct observable $T_{1/2}$ instead, finding a new value of $U = 1.7$ eV after including vibrational corrections to $H(T)$ and $S(T)$. I found that subtle bond distortions in the HS molecule broke the degeneracy of its t_{2g} orbitals, and explicitly tested different d orbital occupations to avoid getting stuck in metastable states which were not the true ground state [43].

With the above improvements, I then recalculated the DFT energies, forces and stresses for the ~ 600 training structures found on-the-fly, the distribution of which is shown in Figure 4(a). The average expansion of Fe-N bonds (Q) in the HS state is noted, with the raw values and relative difference with respect to the LS state all being quantitatively accurate to experiment [60]. An average volume expansion is also observed.

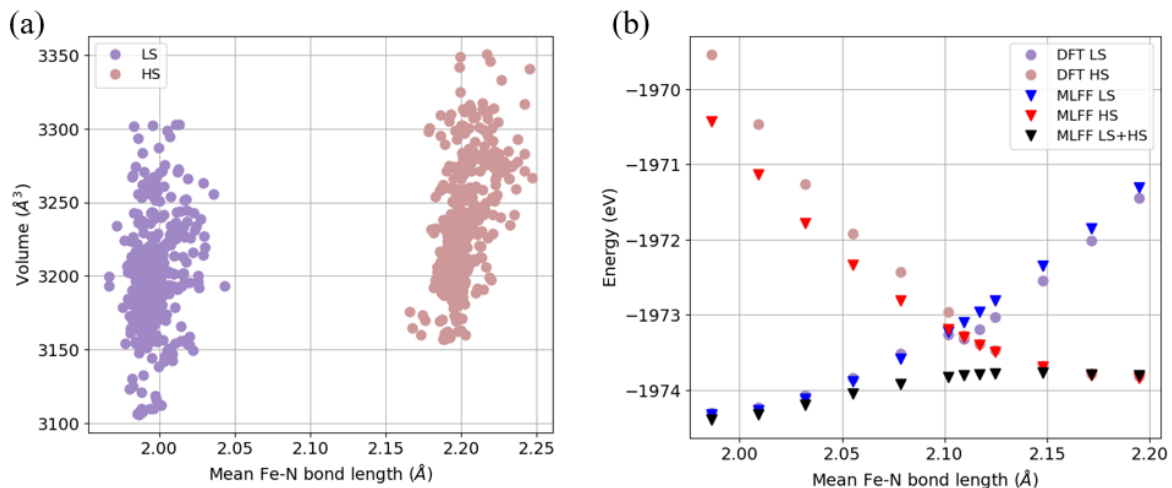


Figure 4: (a) Training data distribution as a function of mean bond length and volume. (b) DFT and MLFF electronic energies along interpolated structures between LS and HS configurations. The combined MLFF (black triangles) is not yet optimal in the intermediate region between spin states and requires further training.

Separate HS and LS MLFFs were subsequently retrained, too. General challenges with this involve trying to avoid overfitting to the training data, particularly when it is tempting to reweight parameters to prioritise, for example, accurate spin state energy differences. Our initial results are promising. Figure 4(b) compares DFT and

MLFF energies (H^{elec}) for a sequence of interpolated structures. The LS and HS MLFFs show excellent agreement with DFT in the structural regime where their given spin state is stable. The black points show a *combined* MLFF, which is trained on the previously sampled LS and HS data combined. Keeping both spin states' data contained to one MLFF would enable continuous MD simulations, making it an appealing prospect. I am in the process of adding new training data, closer to the crossing point at $Q \approx 2.1\text{\AA}$, in order to improve the fit in this region relative to the separate MLFFs. With Q as a tentative order parameter for SCO, I will next investigate the viability of using the combined MLFF to recover $T_{1/2}$ from an MD simulation where the sample is warmed up.

Steps 2 and 3: Multistep materials and uncharacterised structures

Step 2 involves expanding the utility of SCO MLFFs for materials showing more exotic transitions. Cruddas and Powell explored structure-property relationships for some multistep SCO materials, and found that longer-range elastic interactions in ordered lattices are what instigate these kinds of transition dynamics [35]. It will be intriguing from both a fundamental physics perspective and a device design perspective to see how well MLFFs describe the free energy space of these structures, and if more accurate methods like thermodynamic integration are required.

The culmination of the project lies in Step 3, where I will apply the DFT-MLFF-MD pipeline to newly synthesised SCO materials. This is the most challenging of the steps, as I will need to apply some chemical intuition and previous benchmarking experience to decide on a range of U parameters to test. Ultimately, though, this should be a rewarding application of the project, and I will be able to make tailored suggestions to help collaborators rationalise design choices and engineer $T_{1/2}$ at ambient temperature [6]. Some such collaborators are the groups of Professor Cameron Kepert (University of Sydney), Professor Sanjit Konar (Indian Institute of Science, Education, and Research, Bhopal) and Professor Michael Shatruk (Florida State University), all of whom Professor Powell has ongoing collaborations with. Advancing the MLFF methodology so the spin-state can be directly predicted would also be an exciting direction.

4.1 Timeline

Purple sections indicate completion, while blue sections indicate steps yet-to-come.

01/25 - 04/25: Familiarise myself with the fundamental physics of SCO materials. Learn how to submit and run jobs on a HPC cluster. Learn the mechanisms and motivation for performing simple DFT calculations. Learn how to train MLFFs with VASP. Aim to replicate previous student's results for ground-state structures and energies of $[\text{Fe}(\text{ptz})_6](\text{BF}_4)_2$.

05/25 - 07/25: Begin Step 1. Benchmark and apply occupation matrix control for consistent DFT+U energies. Select new U value based on initial MLFF's vibrational corrections near the experimental $T_{1/2}$.

08/25 - 12/25: Train new MLFF on more rigorous DFT+U methodology. Calculate MLFF free energy differences with Phonopy Python package for VASP, and check the DFT-tuned $T_{1/2}$ is recovered.

01/26 - 04/26: Retrain separate spin state MLFFs into a combined FF. Continue on-the-fly training to extend sample space. Perform continuous MD runs to simulate thermal SCO and identify dominant vibrational modes.

05/26 - 08/26: Begin Step 2. Learn the details of elastic models for highly-cooperative, exotic SCO materials. Write Step 1 into a paper and thesis chapter.

09/26 - 04/27: Apply the DFT+U-MLFF-MD pipeline to exotic SCO materials. Trial thermodynamic integration to calculate free energy for multistep materials.

05/27 - 02/28: Step 3. Apply the DFT+U-MLFF-MD pipeline to unpublished SCO materials. Collaborate with experimentalists to suggest effects of different guest molecules or crystal rigidity in tuning $T_{1/2}$. Write Step 2 into a paper and thesis chapter.

03/28 - 06/28: Write Step 3 into a paper. Finalise any other papers still in progress and write the thesis.

5 Outputs

5.1 Conferences and meetings attended

- AIP Condensed Matter & Materials Meeting - Brisbane, Australia (4-7 February 2025)
- Australian Winter School on Computational Chemistry - Kiama, Australia (7-10 July 2025)

5.2 Presented talks and posters

- Group Meeting talks - PhD plan, PhD update
- Poster at AWSCC and SMP Poster Day - ‘What’s shaking in spin crossover?: Machine-learned force fields from a DFT method tuned to lattice vibrations’

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